

B.SC. ENGG. THESIS

A Report on Airline Management System

Submitted to

Department of Computer Science & Engineering

(In partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science & Engineering)



Department of Computer Science & Engineering

Bangladesh University of Business & Technology (BUBT)

Dhaka 1000

March 3, 2021

Acknowledgment

We would like to pay our gratitude to the Almighty Allah who created us with all the abilities to understand analysis and develop the process with patience. We are thankful to our thesis supervisor Nourin Khandaker , Assistant Professor, Computer Science and Engineering Department, Bangladesh University of Business and Technology for his professional guidance and motivation during the work of this thesis which is a major part of it. Without his valuable support and guidance, this thesis could not reach this level of development from our point of view.

We would like to thank all the Faculty members, Department of CSE, Bangladesh University of Business and Technology for their valuable time spend in requirements analysis and evaluation of the thesis work. We would like to express our sincere and warm gratitude to all those who have encouraged us directly, provided mental encouragement and criticized our work in several phases during the development of this thesis and for preparing this thesis indirectly.

Abstract

Airline management systems were first introduced in the late 1950s as relatively simple standalone systems to control flight inventory, maintain flight schedules, seat assignments and aircraft loading. The modern airline management system is comprehensive suite of products to provide a system that assists with a variety of airline management tasks and service customer needs from the time of initial reservation through completion of the flight. One of the most common modes of travel is traveling by air. Customers who wish to travel by air nowadays have a wide variety of airlines and a range of timings to choose from. Nowadays competition is so fierce between airlines that there are lot of discounts and a lot of luxuries given to customers that will give an edge to that particular airline. The World Wide Web has become tremendously popular over the last four years, and currently most of the airlines have made provision for online reservation of their flights. The Internet has become a major resource for people looking for making reservations online without the hassle of meeting travel agents. Our Project intends to serve these purposes. It intends to check all the available airline databases and return a string of results, which can help them in their travel plans. The objective of this project is to create an airline management system where a traveler can request all flight information as per their journey dates. They can get information regarding time, cost, etc all at the same time and place. When the customer calls the Counter Assistant for his/her travel needs, the counter assistant will enter the customer's details (flight requirements) in the system. The system displays all the available airlines, schedules and prices. This system would help the airline to better serve its customers by catering to their needs. The site would use a Database to hold this information as well as the latest pricing and availability information for the airlines. *Air Bangladesh* .

Approval

A Thesis on Online Gadget Store Management System is submitted by

Md. Tajwar Ali Rumman (ID-19202103183)

Tasmia Binte Munir Maisha(ID-19202103195)

Aysha Siddika (ID-19202103165)

Parvas Hossain Piash(ID:19202103200)

under the department of Computer Science and Engineering of Bangladesh University of Business and Technology is accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering

Acronyms List

Contents

<i>Acknowledgment</i>	i
<i>Abstract</i>	ii
<i>Acronyms List</i>	iv
<i>List of Figures</i>	iv
<i>List of Tables</i>	iv
1 Introduction	1
1.1 Introduction	1
1.2 About Project	2
1.3 Objectives of the Project	3
1.4 Purpose of the project	3
1.5 SCOPE OF THE PROJECT	4
1.5.1 Need of Computerisation	4
1.6 Project Overview.	5
1.7 Organization of This Research Report	6
2 Motivation	7
2.1 Existing System	7
2.1.1 Limitation of existing system:	8
2.1.2 Every manager faces lot of minor major problems like:	8
2.2 Proposed System:	9

3	Methodology	10
3.1	Module Description	10
4	System Analysis	12
4.1	An Overview to system analysis	12
4.1.1	WORKING OF THE PROJECT	13
4.1.2	Software Requirement:	13
5	Manual Discussions and results	14
5.1	Discussion	14
5.2	Admin Manual	14
5.2.1	Result Analysis	14
5.3	User Manual	20
5.3.1	Result Analysis	20
6	Future Work and Conclusion	25
6.1	Future Work	25
6.2	Conclusion	26
6.2.1	We can hereby conclude that:	27
6.3	References	27

List of Figures

1.1	Visual Studio.	3
1.2	Shopping through online.	5
2.1	Proposed System.	9
3.1	Er-Diagram.	11
5.1	Login Panel.	15
5.2	Admin Manual Main-menu.	16
5.3	Record New Flights.	17
5.4	Available Flights.	18
5.5	Record New Passengers.	18
5.6	Passenger Records.	19
5.7	Available Tickets	20
5.8	User Manual main-menu.	21
5.9	Flights available for passengers.	22
5.10	Passenger details.	23
5.11	Available tickets	23
5.12	Ticket Booking.	24
6.1	Have a safe journey with Air-Bangladesh.	26

List of Tables

Chapter 1

Introduction

1.1 Introduction

This project on Airline Management System is the automation of registration process of airline system. The system is able to provide much information like passenger's information, criminal's, list of all passengers etc. The system also allows us to add records when a passenger reserves a ticket. For data storage and retrieval we use the file-handling facility of C Language. It enables us to add any number of records in our database. But for intrinsic nature of file handling, the retrieval process is slow when we Search a particular record in the database, because record is searched sequentially. Assistance is provided to the user at each and every step so that no problem is faced during using it. Further the details of every process and the user manuals attached in the report make it very easy to understand. Every possible care has been taken to make the software and the report clear, simple and error free which makes it so special and one of its kind. The Airline Management System is a database-driven desktop application written in C and .NET framework. Its user friendly nature and in-built documentation, complication, error detection, binding facilities and interaction with other software packages make it most powerful tool for software development. The project's major goal is to manage the details of Airlines Ticket,Flights,Customer,Booking.

1.2 About Project

The project named “Airline Management System” is written in Visual Studio(2022), mainly because of it’s suitability for this type of application. Its user friendly nature and in-built documentation, compilation, error detection, binding facilities and interaction with other software packages make it most powerful tool for software development. Moreover .Visual Studio(2022) consists of all the technologies that help in creating and running robust, scale able and distributed packages. In Visual Studio, Java is a general-purpose object-oriented programming language, and is intended to be an improved C with object capabilities Assistance is provided to the user at each and every step so that no problem is faced during using it. Further the details of every process and the user manuals attached in the report make it very easy to understand. Every possible care has been taken to make the software and the report clear, simple and error free which makes it so special and one of its kind.



Figure 1.1: Visual Studio.

1.3 Objectives of the Project

- To provide some amount of automation in airlines management.
- To help airlines system in making their business more efficient.
- An added attraction for their potential customers.
- It will also show the attitude of the management that they are aware to the newly introduced technology and ready to adopt them.

1.4 Purpose of the project

Electronically handling of flight's record to enhance the accuracy, flexibility, reliability and to remove the human's error.

- An airline provides air transport services for passengers, generally with a recognized operating
- To provide accurate information about the addition, deletion and modified movie's record.

- To provide, efficient, accurate, reliable, fast, and robust structure that can handle any number of passenger's transactions.

1.5 SCOPE OF THE PROJECT

This project on Airline Management System is the automation of registration process of airlines system. The system is able to provide much information like passenger's information, criminal's, list of all passengers etc. The system also allows us to add records when a passenger reserves a ticket. For data storage and retrieval we use the Visual Studio, Azure-Database facility of Java Language. It enables us to add any number of records in our database. But for intrinsic nature of file handling, the retrieval process is slow when we search a particular record in the database, because record is searched sequentially.

1.5.1 Need of Computerisation

A few factors that directs us to develop a new system are given below :-

- 1) Faster System
- 2) Accuracy
- 3) Reliability
- 4) Informative
- 5) Reservations and cancellations from any where to any place

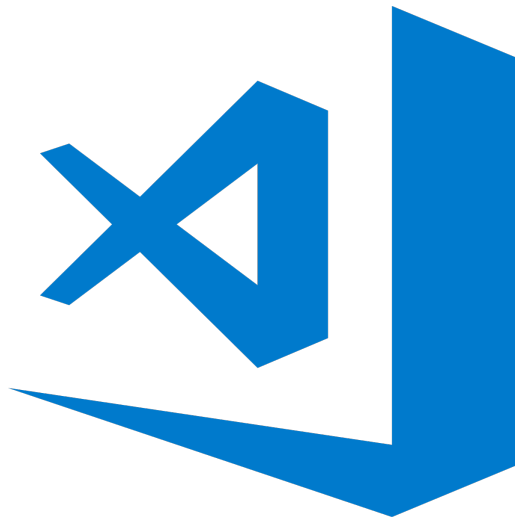


Figure 1.2: Shopping through online.

1.6 Project Overview.

- * Database and database systems have become an essential component of everyday life in modern society. In the course of a day, most of us encounter several activities that involve some interaction with the database. For example, if we go to the bank to deposit or withdraw funds or if we make a Hotel or Airline Reservation, chances are that our activities will involve someone accessing a database.
- * The above interactions are examples of what we may call traditional database applications, where most of the application that is stored and accessed is either textual or numeric. In our project we will concentrate on this aspect of computer application.
- * There are several ways to implement databases. Some of them are file handling mechanism, relational database, object-relational database or object-oriented databases. In our project we will use file-handling feature provided by C++ Language.

- * This program shows you an insight into the management process of reservation in Airline Management system. The whole process of Airline Management System is shown with the help of this project. It provides facility to add/Modify/Delete/search Airline Management details. and provide facility to view the list of Team .
- * Facility to view the list of Team .

1.7 Organization of This Research Report

The rest of the book is organized in the following way. In Chapter 1, we will show the background and related research studies. After that,

- **In Chapter 2**, Motivation of Airline Management. Limitations of existing system. Merits of proposed System.
- **In Chapter 3**, From this chapter we will get the brief idea of our methodology . We will discuss about the structure of our project.
- **In Chapter 4**, We will find out about system analysis,software and hardware requirements.
- **In Chapter 5**, The overview of the Admin and User manual system. We will learn more about the information load of the Airline management. Moreover,this chapter shows the results and the project outputs. We can also find the analysis of the result.
- **In Chapter 6**, In this chapter we will find the the the conclusion of the whole report and our future intentions. What steps are we going to take in the very future, these information are given here.

Chapter 2

Motivation

Before making this application, we assumed that an airline which had recently started its operation found it very difficult to handle their customers. It was due to their great customer service and efficient handling of daily operations that their customer base started growing and in a day, they started to handle a lot of customer requests. The problem is that in a manual airline record keeping system, excessive staff employment is required, an extremely time-consuming process is involved, inconveniences to both customers as well as to the manager. Slowly, slowly the count of such customers started to grow very rapidly and the airline employees had to devote their maximum time in handling such customers. Slowly, an airline started losing its important or gold customers due to poor response times by the employees and they even started losing those customers whose requests could not be fulfilled. After this, the management decided to install a system that can effectively and efficiently service the request of such customers and can take the corresponding work of its employees who were overburdened with such tasks. This action was a step towards serving important or fresh customers with a minimum possible and improve the response times and efficiency of an airline employees.

Objective of this software is to simplify the employee record using computers

2.1 Existing System

The system is very time consuming and lazy. This system is more prone to errors and sometimes the approach to various problems is unstructured. If any old data or information is

to be fetched then it is a great problem for user to get the information in short span of time as to get information from files is not an easy task. As everything is done manually, so if any record is misplaced then agency has to take full responsibility.

2.1.1 Limitation of existing system:

The earlier experiences have shown that manual monitoring of employee enquiries about their loans, conveyance, etc. Often fails to achieve the desired targets, mainly because of the following reasons:

- Much time required in giving correct information.
- Less reliability and maintainability of data.
- Secrecy of information may not be maintained due to visible facts on paper.
- Manual procedure of providing information is not reliable.

2.1.2 Every manager faces lot of minor major problems like:

- 1) Maintaining database.
- 2) Record entry.
- 3) Searching duplicate records.
- 4) Searching updating records.

An object oriented system draws upon class definitions that are derived from the analysis model. Some of the definitions will have to be built from scratch but many other can be reused if appropriate design patterns are recognized. Object oriented design establishes a design blueprint that enables a software engineer to the object oriented architecture in a manner that maximized reuse, thereby improving development speed and product quality.

2.2 Proposed System:

The proposed system is computer based, user friendly, and easy to maintain. It makes safely storing of records easy and for a very long period of time. It would significantly improve the quality of work in the airport. The time spent in processing the above mentioned queries would significantly reduce. The proposed system provides free, easy and efficient management of the day-to-day activities of the passenger's in airline so that the manual work can be reduced and even minute details can be accessed easily.



Figure 2.1: Proposed System.

Chapter 3

Methodology

3.1 Module Description

In this management system there is a main Admin panel. The admin can manipulate both the admin and the user panel.

On the other hand, AIRLINE MANAGEMENT SYSTEM is basically a menu driven program. This kind of format is designed considering the user requirements. This is to provide an easy and faster method of operation to the user. Firstly the user has to sign up and then login through that id and password and then the booking process start in which first customer mention its details which will be stored in database , then further it will be carried on to selecting the flight details by selecting origin and destination, the seat will be booked for the customer and also this software allows the customer to book n number of tickets from one email id. and then finally the payment part will be shown after successful payment seat will be booked.

We have implemented validation at 03 points in the system as:

1. The first point is where the user enters his account pin no. This is the most important part of our application because the information which would be fetched is displayed to the user is confidential and it should be displayed only after proper authentication. So, for security reasons, we have given only one chance to the user to enter his pin correctly. If he does not, the system issues a warning through a proper message and exits. The user then again has to swipe his card, enter his account pin no. to view the account details or undertake any

account activity.

2. The second point where this is implemented is the menu where the user chooses from a list of options to process his requests. Since this is a menu-driven program, we expect from the user to input correct option. But if, for some reason, the user is unable to enter it correctly, we flash a message which requests the user to enter a correct option.

3. The third point where we have taken care of user input is the place where user wants to withdraw money from his account. Ideally, the user should not enter the withdrawal amount greater than his total amount and if mistakenly he does, the system flashes a user message and inform him about the same.

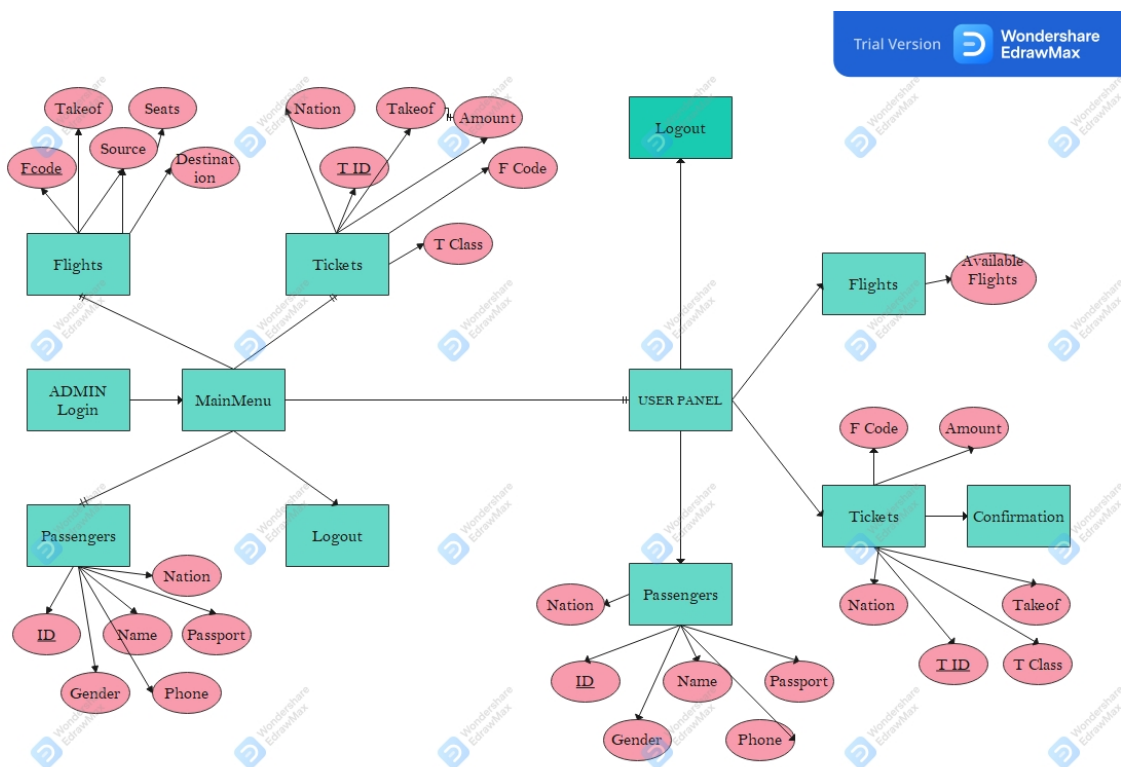


Figure 3.1: Er-Diagram.

Chapter 4

System Analysis

The analysis model must achieve three primary objectives:- 1. To describe the requirements of the customer. 2. To establish a basis for the creation of a software design. 3. To define a set of requirements that can be validated once software is built.

4.1 An Overview to system analysis

The system analysis phase is considered to be one of the most important phases in the system development life cycle. It is immensely important that the software developer make through study of the existing system. Thorough study of the system is made and need i.e. features that are critical to system success and users wants (i.e. features that would be good but not essential) are brought out. The study will enable the developer to know the intricacies of the existing system. Requirement analysis is done in order to understand the problem which the S/W system is to solve e.g., the problem could be automating the existing manual system or developing a completely new automated system or a combination of the two. For large systems having a large number of features and the need to perform many different tasks, understanding the requirement of the system is a major task. The emphasis in requirement analysis is on identifying what is needed from the system, and not how the system achieves its goal. The main objective behind any business organization is to maximize its profit besides maintaining quality and strategic norms. This can be achieved by improving the efficiency of the system by providing more facilities using automation, by adopting faster data access, proper

communication. , whereas the main objective behind automation is not only to maximize profit but also to take care of passenger's interest by providing them better facilities. The most important objective behind automation is to minimize Paper Work. Paper Work/Registers are replaced by a Centralized Data Bank, which is well equipped to store / provide information as and when required. Data Bank also helps speed up the

communication between various depts. outside agencies, as there is no need of making request against different departments for a specific data and to wait for it for a long period. This also improves the efficiency as it saves time and human resources. By making the manual system computerized, we can ensure complete utilization of our existing resources. Automation helps in generating the reports / information in a consistent way, which saves time and labour if done manually. In this project we have used Rapid Application Development (RAD) model. RAD is an incremental software development process model that emphasizes an extremely short development cycle.

4.1.1 WORKING OF THE PROJECT

- User can view record about flight by selecting option 1 from the main menu.
- User can reserve the seat for view the flight. by selecting option 2.
- User can also cancel the reserved ticket for flight by selecting option 3.
- Admin can collect total amount. by selecting option 4.

4.1.2 Software Requirement:

Platform Support: Windows 10 or advanced.

Language: .NET, C

Software: Visual Studio 2022

Database: MySQL Server(Azure)

Chapter 5

Manual Discussions and results

5.1 Discussion

In this chapter, there will be discuss about our project. The reason behind the development of this project is to make Air journey easier for the people. We have created both the Admin Manual and the User manual of the project It was really satisfying when the project of ours was a success and the output came. The hardship of our team really payed off. Now few outputs of our code are provided through pictures below.

5.2 Admin Manual

5.2.1 Result Analysis

- Login/Sign up.
- Main Menu.
- Flights.
- passengers.
- Tickets.
- Log out/ Exit.

Output:

Admin Panel



The image shows a web application interface for the Air Bangladesh Admin Panel. It features a dark blue sidebar on the left with the Air Bangladesh logo (a stylized blue and white wing with a small airplane) and the tagline "Let Your Dreams Take Flight!". The main content area is light gray and contains the "Air Bangladesh" logo at the top. Below the logo, there are input fields for "Admin Id:" (containing "111") and "Password:" (containing "***"). At the bottom of the form are two buttons: "Login" and "Sign up". A dark blue vertical bar on the right side of the panel has a white "X" icon at the top.

Figure 5.1: Login Panel.

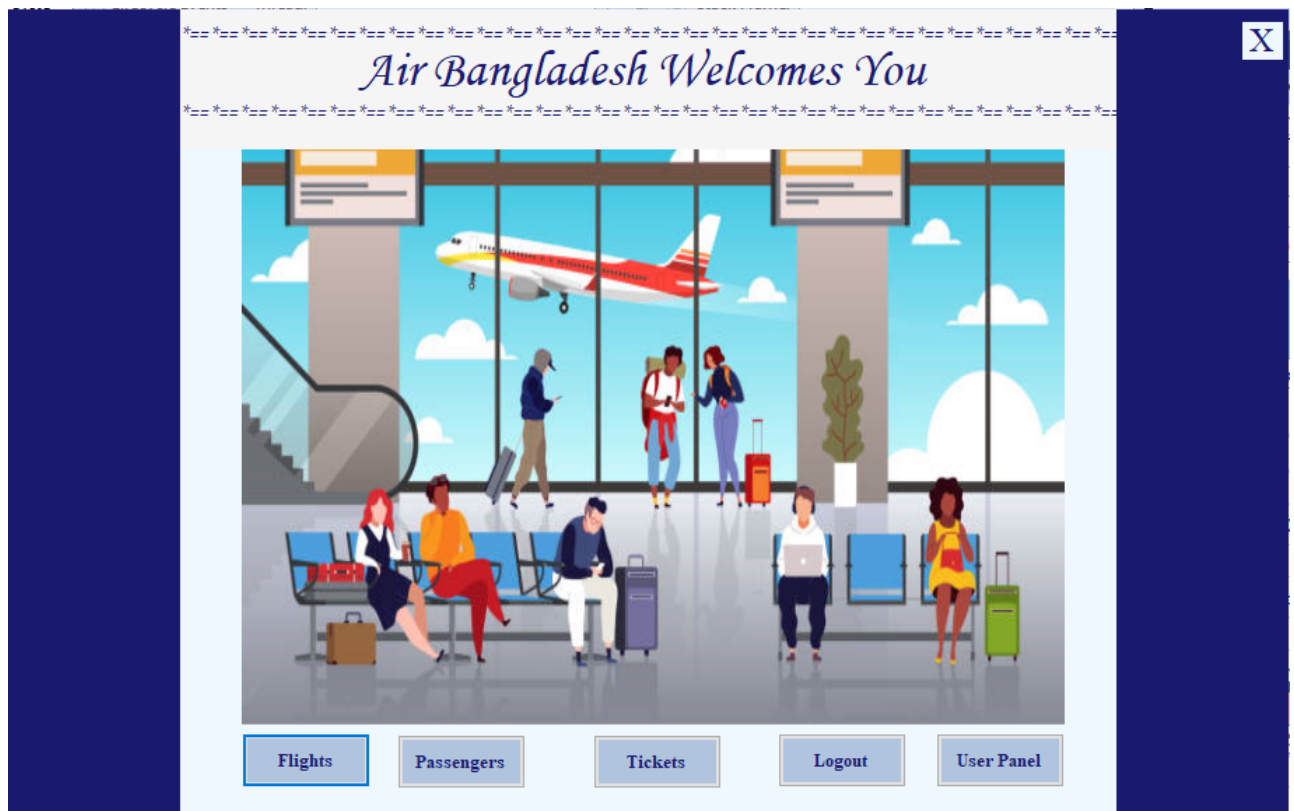


Figure 5.2: Admin Manual Main-menu.

X

Record of New Flights

Flight Code

Source

Destination

Takeof Date

Number of seats



Figure 5.3: Record New Flights.

X

View scheduled Flights

Flight Code

Source

Destination

Number of seats

Takeof Date

	Fcode	Fsrc	FDest	FDate	FCap
▶	A71	Japan, Tokyo	Bangladesh, Dh...	Saturday, April 2...	500
	A73	China, Beijing	Bangladesh, Dh...	Sunday, April 3, ...	250
	AirBD04	Bangladesh, Dh...	India, New Delhi	Thursday, March...	300
	AirBD05	Bangladesh, Dh...	Pakistan, Islama...	Saturday, March...	500
	Sky006	Bangladesh, Dh...	Japan, Tokyo	Tuesday, March ...	500
✱					

Figure 5.4: Available Flights.

===== X

Record Passengers

=====

Passenger ID

7

Passenger Name

Plabon

Passport No.

707

Passenger Address

Mohammadpur

Nationality

Canada

Gender

Male

Phone

01874955196

Record

Reset

View

Back



View Passengers

Passenger ID

Passenger Name

Passport No.

Passenger Address

Nationality

Gender

Phone

Update Delete Reset Back

	PassId	PassName	Passport	PassAddress	PassNation	PassGender	PassPhone
	1	bushra	12	narayongonj	Pakistan	Female	121231
	2	cicily	232	shiyalbari	USE	Female	123213
	3	Rumman	555	Kaziparan, Mirpur	Bangladesh	Male	01933954158
	4	Plabon	141	Mohammadpur	Bangladesh	Male	01751358711
	5	Maisha	955	Monipu, Mirpur	Bangladesh	Female	01862953154
▶▶							

Figure 5.6: Passenger Records.

Ticket Booking

Ticket Id

Flight Code

Ticket Class

Nationality

Amount

Takeof Date

Book Reset Cancel

	Tid	TFcode	Tnation	Tclass	Tamount	Tdate
▶	1	AirBD05	Bangladesh	Premium-> 15000/-	Premium Econo...	Saturday, Marc...
	11	AirBD05	Bangladesh	Premium-> 15000/-	Premium Econo...	Saturday, Marc...
*						

Ticket details successfully recorded.
OK

Back

Figure 5.7: Available Tickets .

5.3 User Manual

5.3.1 Result Analysis

- Main Menu.
- Flights.
- passengers.
- Tickets.
- Log out/ Exit.

Output:

User Panel

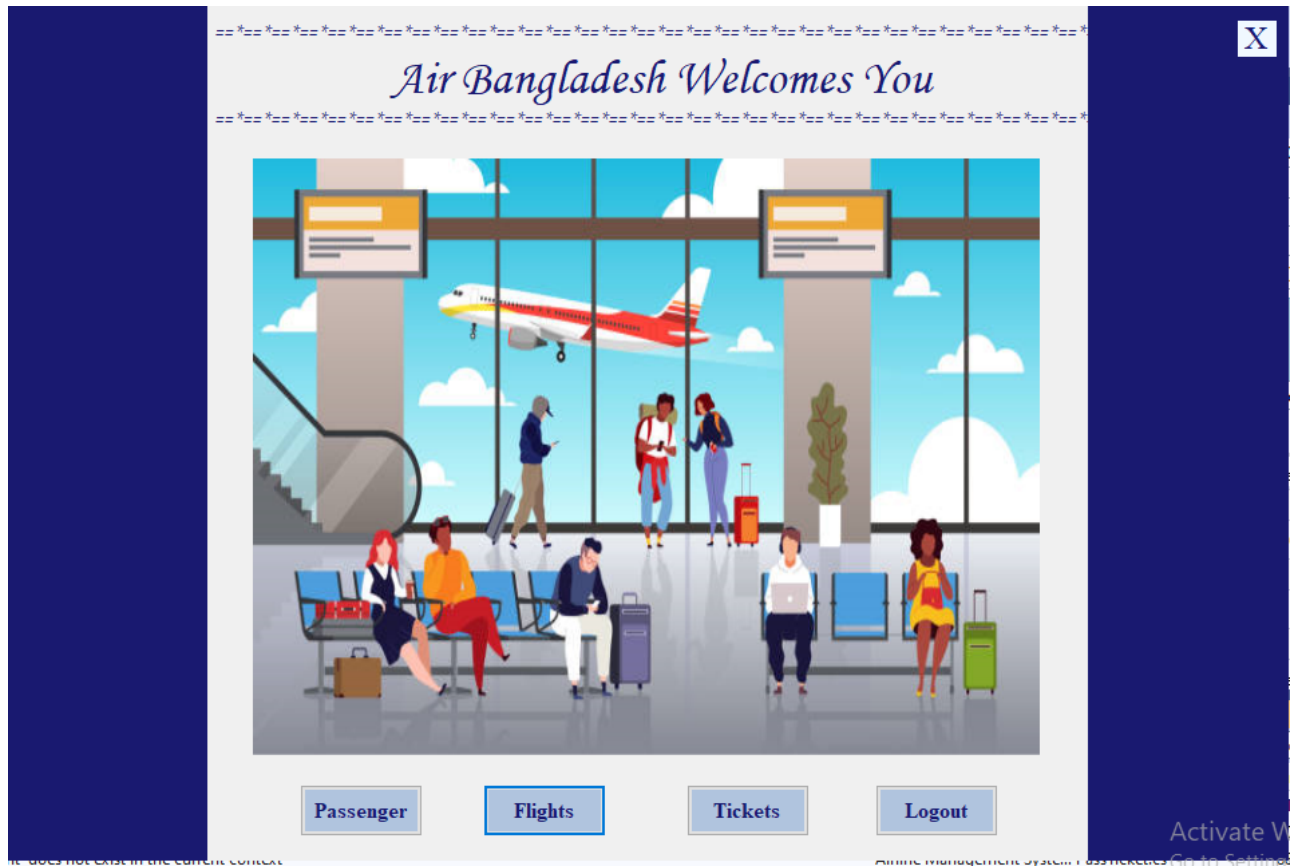


Figure 5.8: User Manual main-menu.

X

Flights



Book ticket: Tickets

Available Flights

	Fcode	Fsrc	FDest	FDate	FCap
▶	A71	Japan, Tokyo	Bangladesh, Dha...	Saturday, April 2, ...	500
	A73	China, Beijing	Bangladesh, Dha...	Sunday, April 3, 2...	250
	AirBD04	Bangladesh, Dha...	India, New Delhi	Thursday, March ...	300
	AirBD05	Bangladesh, Dha...	Pakistan, Islamab...	Saturday, March ...	500
	Sky006	Bangladesh, Dha...	Japan, Tokyo	Tuesday, March ...	500
*					

Back

Figure 5.9: Flights available for passengers.

X

Passengers Information

Passenger ID

Passenger Name

Passport No.

Passenger Address

Nationality

Gender

Phone

Record
Reset
Back



Figure 5.10: Passenger details.

X

Ticket Booking

Ticket Id

Flight Code

Ticket Class

Amount

Quantity

Takeoff Date

[Book](#) [Reset](#)

[View](#)

[Back](#)



Total cost

	Tid	TFcode	Tnation	Tamount	Tclass	Tdate
▶	1	AirBD05	Bangladesh	Premium-> 15000/-	Premium Economy	Saturday, March 26, 2...
	2	Sky009	India	Firstclass-> 12000/-	First class	Wednesday, March 2...
	3	A71	Bangladesh	secondclass->10000/-	second class	Thursday, April 7, 2022
*						

Figure 5.11: Available tickets .

Ticket Booking

Ticket Id

Flight Code

Ticket Class

Amount

Quantity

Takeof Date



Total cost

	Tid	TFcode	Tnatio	Class	Tdate
▶	1	AirBD05	Banglade	um Economy	Saturday, March 26, 2...
	2	Sky009	India	lass	Wednesday, March 2...
	3	A71	Banglade	nd class	Thursday, April 7, 2022
*					

Figure 5.12: Ticket Booking.

Chapter 6

Future Work and Conclusion

6.1 Future Work

Developing this application made we learn many things about C, .NET and Database management systems and developing desktop applications by integrating databases. Future Scope: • This project can be upgraded by adding more options such as Ticket editing and more admin operations.

- Payment options and document checking such as ID proofs can be added.
- Applications can be upgraded by improving performance as per user k .Different database connection procedure is complex. Here we used only Microsoft SQL Sever.



Figure 6.1: Have a safe journey with Air-Bangladesh.

6.2 Conclusion

Scope of this project is to replace the manual system with the new advanced computerized system. User does not need much training to use this software, as this software is very user friendly and easy to use. It replaces all the paper work also. In this software we can store thousands of records. It replaces all the calculation works also as it automatically calculate and print the bill. Some main advantages are:- Customer numbers are generated automatically by system itself. If any invalid entry done by the users which is nature of human being but checks by the machine so this software gives the error message to the users to indicate about the invalid entry. While transfer of any customer such as incoming customer, it will update both the file and incoming customer so that if we want to know number of incoming customer, it will be very easily for us.

6.2.1 We can hereby conclude that:

-
- The system effectively automated the functions involved in the processes being handled manually before.
- The cost benefit analysis shows that the system was quite successful in saving costs for the bank generate equivalently huge benefits
- The system is secure scale able. The system design has been done keeping user- friendliness and efficiency in mind.

6.3 References

1. www.google.com
2. www.youtube.com
3. www.w3school.com
4. <https://www.javatpoint.com/>
5. <https://dev.mysql.com/doc>

Bibliography